25900248 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) Regular EXAMINATION October 2025

Third SEMESTER

Discipline Specific Core Course (DSC) - MG3DSCBTG200 - BIOPHYSICS AND INSTRUMENTATION

(2024 ADMISSION ONWARDS)

Duration: 1 Hour 30 Minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

One word or sentence

Answer any **10** questions

Each question carries **2** marks

1. Give the name for a common Adsorbent? [K] / [CO1]
2. Recall two methods of paper chromatography. [K] / [CO3]
3. State the principle behind dialysis. [U] / [CO3]
4. Enumerate any two chemical properties of water. [K] / [CO1]
5. Recall which amino acid shows restricted angles in Ramachandran plot. [K] / [CO2]
6. Name one isotope used in autoradiography. [K] / [CO4]
7. Identify the type of molecules separated by Gel Permeation Chromatography. [U] / [CO3]
8. State the zeroth law of thermodynamics. [K] / [CO1]

9. Comment on the primary advantage of using HPLC over Gas chromatography? [U] / [CO3]

10. What is required in electron microscopy to prevent scattering of electrons? [K] / [CO4]

11. Name the source of IR radiation. [K] / [CO4]

12. State why GC-rich DNA has a higher melting temperature. [U] / [CO2]

[2x10 = 20]

Part B

Short Essay

Answer any **5** questions

Each question carries **4** marks

13. Describe the tertiary structure of protein and list the types of bonds that stabilize it. [U] / [CO2]

14. Explain enthalpy and entropy including examples of its application in biology. [K] / [CO1]

15. Recall the advantage of PAGE over AGE. [K] / [CO3]

16. State the difference between dispersion and scattering (any 4 points). [K] / [CO1]

17. Explain the structure of four-stranded DNA (G-quadruplex) with the help of a diagram. [U] / [CO2]

18. Define and explain the concept of sedimentation coeffecient and its unit [K] / [CO3]

19. Mention and explain four applications of absorption in daily life. [K] / [CO1]

[4x5 = 20]

Part C

Essay type questions

Answer any **1** questions

Each question carries **10** marks

20. Define a biosensor and outline its principle and basic working mechanism. [K] / [CO4]

21. Explain in detail the concept of diffusion and importance in biological context. [K] / [CO1]

[10x1 = 10]